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Introduction to Database Systems and Evolution of Database Technology

From Data to Databases: The Journey of Data Management



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Learning Outcomes

- By the end of this lecture, students will be able to:
- Understand the need for databases.
- Differentiate between File Systems and DBMS.
- Define Database and DBMS.
- Explain major components of a database system.
- Understand database applications.
- Appreciate the evolution from Relational Databases to NoSQL and Cloud Databases.



Why Do We Need Databases?

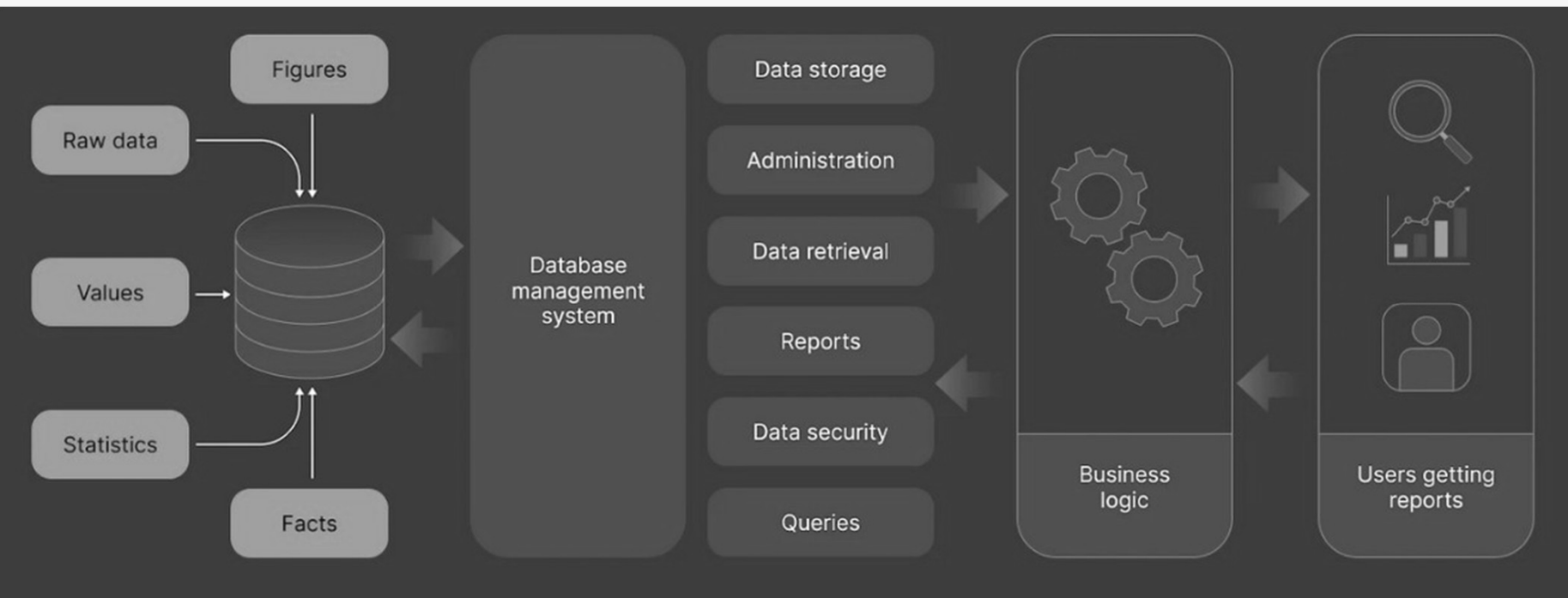
- Before Databases Organizations stored information using
 - Paper files
 - Registers
 - Flat files
 - Text files
 - Excel sheets
- Problems
 - Data duplication
 - Difficult searching
 - Slow processing
 - No data sharing
 - Security issues
 - Inconsistent data
 - No backup mechanism
- Example
 - A college stores student information separately in Admission Office, Library, Hostel, Examination Cell
 - If a student's phone number changes, it must be updated in all places.
 - Failure causes data inconsistency.



The Database Management System



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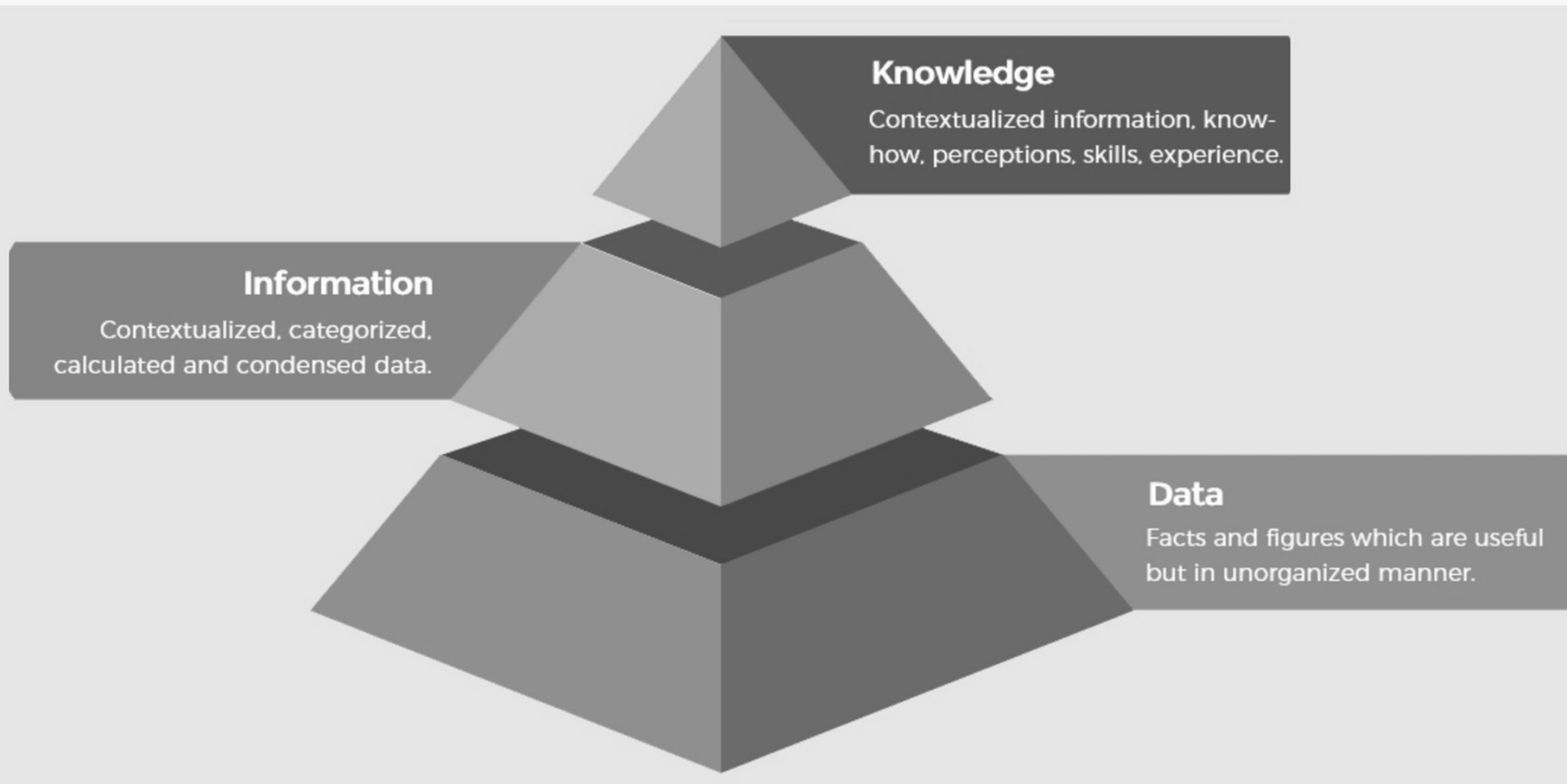
What is Data?

- **Data**
 - Data refers to raw facts, observations, numbers or symbols that have not yet been processed.
- **Example of Data**
 - 101
 - Rahul
 - 82
- **Information**
 - Information is processed and organized data that is meaningful and useful.
- **Example of Information**
 - Student Rahul scored 82 marks.
- **Knowledge**
 - Knowledge is the interpretation of information for decision making.
- **Example of Knowledge**
 - Students scoring below 40 require remedial classes.



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Data, Information, Knowledge



What is a Database?

- **Definition**

- A Database is an organized collection of logically related data stored electronically so that it can be easily accessed, managed and updated.

- **Characteristics**

- Organized
 - Persistent
 - Shared
 - Integrated
 - Secure
 - Efficient retrieval

- **Example**

- University Database
 - Contains
 - Students
 - Faculty
 - Departments
 - Courses
 - Results
 - Fees



What is DBMS?

- **Definition**
 - A Database Management System (DBMS) is software that enables users to create, store, retrieve, update and manage databases efficiently.
- **Responsibilities**
 - Data storage
 - Query processing
 - Security
 - Backup
 - Recovery
 - Concurrency control
- **Examples**
 - Oracle
 - MySQL
 - PostgreSQL
 - SQL Server
 - SQLite



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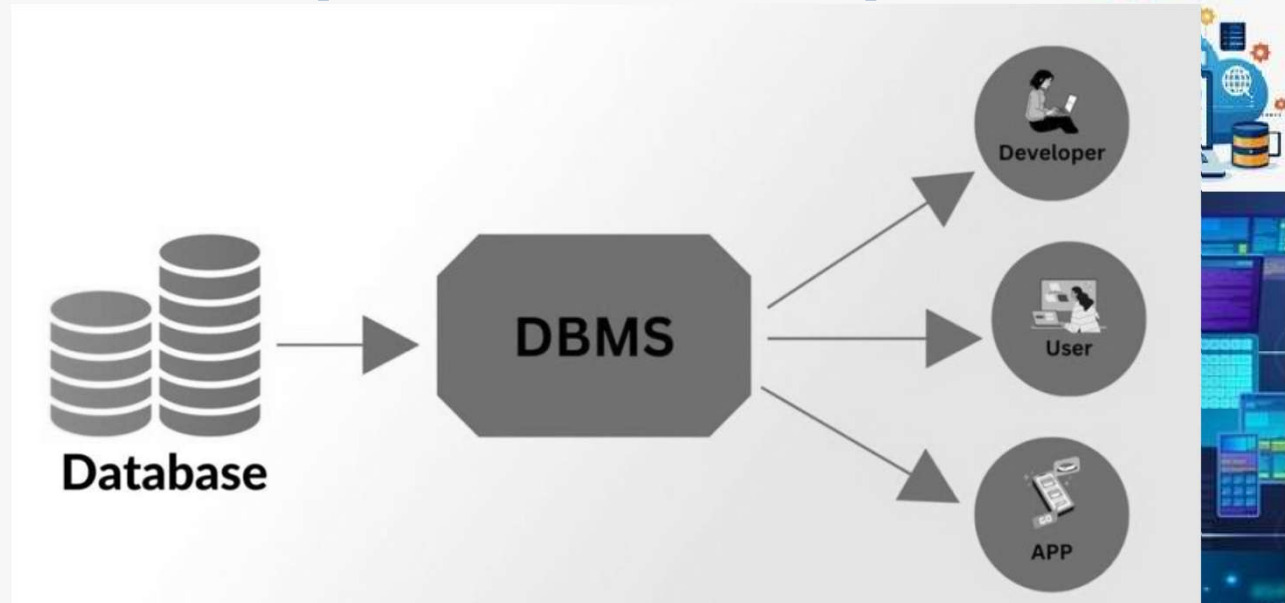
DBMS Architecture (Basic View)

- **Components**

- Users
- Applications
- DBMS
- Database
- Operating System
- Storage

- **Example**

- **ATM → Banking Application
→ Oracle Database → Bank
Records**



Advantages of DBMS

- **Major Advantages**

- ✓ Reduced redundancy
- ✓ Improved consistency
- ✓ Data sharing
- ✓ Security
- ✓ Backup & Recovery
- ✓ Integrity
- ✓ Concurrent access
- ✓ Better decision making

- **Example**

- Amazon stores millions of products in a database instead of thousands of spreadsheets.



File System vs DBMS

- File System

- Separate files
- High redundancy
- Difficult search
- Weak security
- No concurrency
- Manual backup
- Poor consistency
- Example

- Hospital

- File System

- *Separate files for OPD, Pharmacy and Billing.*

- DBMS

- *One integrated database.*

- DBMS

- Centralized database

- Minimal redundancy

- Fast querying

- Strong security

- Multi-user support

- Automatic backup

- High consistency



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Real-Life Applications of Databases

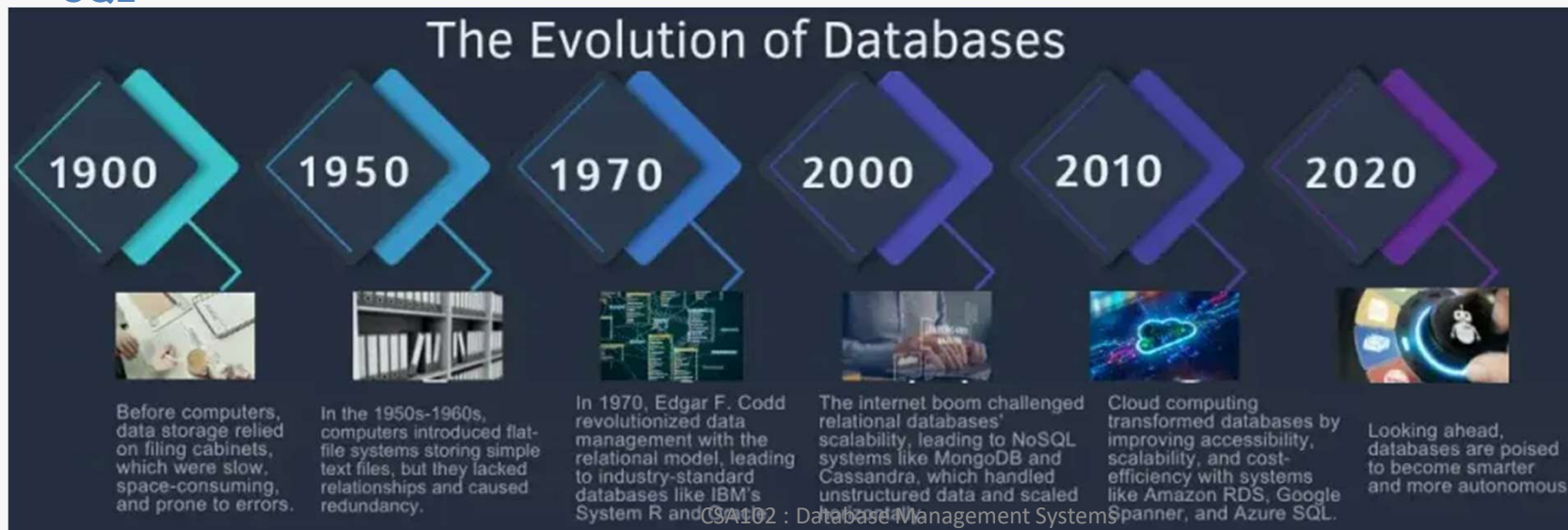
- **Banking**
 - Customer accounts
 - ATM
 - Transactions
- **Healthcare**
 - Patients
 - Reports
 - Prescriptions
- **E-Commerce**
 - Products
 - Orders
 - Payments
- **Education**
 - Student records
 - Attendance
 - Exams
- **Social Media**
 - Profiles
 - Posts
 - Followers
- **Transportation**
 - Railway Reservation
 - Airline Booking



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Evolution of Database Technology

- 1. File Processing Systems
 - Separate files
 - Manual programming
- 2. Hierarchical Database
 - Tree structure
- 3. Network Database
 - Graph-like structure
- 4. Relational Database
 - Tables
 - SQL
- 5. Object-Oriented Database
 - Stores objects directly
- 6. NoSQL Database
 - Document
 - Key-value
 - Graph
 - Column-family
- 7. Cloud Database
 - Managed over Internet



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Introduction to NoSQL (Overview Only)

- Why NoSQL?
 - Modern applications generate
 - Huge data
 - Unstructured data
 - High-speed data
- Traditional RDBMS may not always be sufficient.
- Types
 - Document Database
 - *Example: MongoDB*
 - Key-Value Database
 - *Example: Redis*
 - Column Database
 - *Example: Cassandra*
 - Graph Database
 - *Example: Neo4j*
- Important Note
 - NoSQL does not replace SQL It complements relational databases.



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SQL vs NoSQL

- SQL
- Tables
- Fixed Schema
- ACID
- Structured Data
- Complex joins

- Examples

- SQL
 - Oracle
 - PostgreSQL
 - MySQL

- NoSQL
 - MongoDB
 - Cassandra
 - Redis

NoSQL

Documents/Key-value/Graph

Flexible Schema

BASE (generally)

Semi/Unstructured Data

High scalability



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Summary

- We Learned
- ✓ Data
- ✓ Information
- ✓ Database
- ✓ DBMS
- ✓ Need for DBMS
- ✓ File System vs DBMS
- ✓ Applications
- ✓ Evolution of Databases
- ✓ Introduction to NoSQL & Cloud Databases



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Learning Reflection

- 1. Why is Excel not considered a DBMS?
- 2. Name five applications where databases are essential.
- 3. Why do companies like Amazon and Google need databases?
- 4. Can MongoDB completely replace MySQL? Why or why not?



Homework

- Explain the need for a Database Management System (DBMS) and compare it with the traditional file system. Discuss any four major advantages of DBMS and briefly explain how databases have evolved from file-processing systems to modern relational and NoSQL databases.



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Thanks!

📌 Heads Up!

These slides are just a **guide** — not the full story!

To really learn, make sure to read your book, and explore other resources too.